DEEP RESEARCH Prompt:

Scientist Information

- **Name:** Lincoln Stein
- **Dates:** 1960
- **Era/Field:** Biological data integration and visualization
- **Course context:** BIOL 3030 intro biochemistry audience; collaborative class eBook; final narrative will be ~500 words written by the student

Your Mission

Collect high-signal raw materials (facts, quotes, scenes, human details) that will allow me to write a short narrative profile. Do not choose or assume a narrative lens at the start. First do unbiased research. Only after compiling evidence, recommend the best narrative lens using the scoring rubric at the end.

Core Constraints (Non-Negotiable)

- **No finished prose profile.** Output research materials only.
- **Citations for every meaningful claim.** If you cannot cite it, label it as unverified and treat it as suspect.
- **Prefer primary sources.** Letters, interviews, notebooks, Nobel lectures, oral histories, autobiographies.
- **Myth defense:** Flag details that are widely repeated but weakly sourced (especially precise timestamps, dramatic dialogue, too-perfect anecdotes).
- **Accuracy over drama:** Vividness only counts if it is documented.

What You Need to Find

1) Anchor Scene Candidates (Find 3–6)

Identify multiple specific events that could plausibly serve as the backbone of a narrative. For each scene, capture:

- **What happened:** discovery, decision, conflict, setback, collaboration, turning point, institutional barrier
- **When and where:** date (or best estimate), institution/lab/city, conference, wartime context, etc.
- **Who was involved:** key people present or affected
- **Concrete texture:** workspace, instrument/method, letter content, meeting setting, observed behavior, reactions
- **Why it mattered:** scientific impact, career impact, human impact
- **Evidence quality:** primary vs secondary; how direct the documentation is
- **Myth risk:** is this a “too perfect” story that may be repeated without strong sourcing?

Output each scene as a card:

SCENE #1: Entering Bioinformatics "Through the Back Door"

- **Summary**: Lincoln Stein's transition from clinical pathology into computational biology.
- **Date/Place:** Summer 1992; Whitehead Institute/MIT Center for Genome Research, Cambridge, MA.
- **People**: Lincoln Stein; MIT/Whitehead informatics staff.
- **Concrete details**: Stein later wrote "I slipped into bioinformatics through the back door… while completing my pathology residency … I became frustrated with my inability to combine … biological research with … computing … so when I was offered the opportunity to spend a summer working with the software group … I jumped at it."
- **Why it matters**: Marks a pivotal career shift from clinical pathology toward computational biology, setting the stage for his influence on genome data infrastructure.
- Evidence quality: Secondary; first-person retrospective published online.
- **Myth risk:** Medium — the phrasing "back door" is memorable, but based on Stein's own retrospective statement.
- **Sources:**
    - https://vectorinstitute.ai/team/lincoln-stein/?utm_source=chatgpt.com
    - https://scienceblogs.com/digitalbio/2007/06/19/to-bioinformatics-students-eve?utm_source=chatgpt.com

SCENE #2: Developing the First Physical Clone Map of the Human Genome

- **Summary**: Lincoln Stein's transition from clinical pathology into computational biology.
- **Date/Place:** Summer 1992; Whitehead Institute/MIT Center for Genome Research, Cambridge, MA.
- **People**: Lincoln Stein; MIT/Whitehead informatics staff.
- **Concrete details**: Stein later wrote "I slipped into bioinformatics through the back door… while completing my pathology residency … I became frustrated with my inability to combine … biological research with … computing … so when I was offered the opportunity to spend a summer working with the software group … I jumped at it."
- **Why it matters**: Marks a pivotal career shift from clinical pathology toward computational biology, setting the stage for his influence on genome data infrastructure.
- Evidence quality: Secondary; first-person retrospective published online.
- **Myth risk:** Medium — the phrasing "back door" is memorable, but based on Stein's own retrospective statement.
- **Sources:**
    - https://moleculargenetics.utoronto.ca/news/medgen-faculty-spotlight-dr-lincoln-stein?utm_source=chatgpt.com

SCENE #3: *WormBase Gets Launched*

- **Summary:** Creation/leadership of WormBase, a model-organism genomic database.
- **Date/Place:** Late 1990s–early 2000s; Cold Spring Harbor Laboratory, NY (and broader community).
- **People:** Stein; C. elegans research community.
- **Concrete details:** Stein "led the creation and development of Wormbase," a community base for *Caenorhabditis elegans* genomic data.
- **Why it matters:** WormBase became a prototype for organized biological data sharing, influencing later databases such as Reactome.
- **Evidence quality:** Secondary organizational sources
- **Myth risk:** *Low* — factual project leadership, well documented.
- **Sources:**
    - https://bcc2020.sched.com/speaker/lincoln.stein?utm_source=chatgpt.com

- https://bioinformatics.ca/instructors/lincoln-stein/?utm_source=chatgpt.com

SCENE #4: *Reactome Launch*

- **Summary:** Development of Reactome, a curated knowledgebase of biological pathways.
- **Date/Place:** 2003–2011 (project inception to launch); Ontario Institute for Cancer Research and international partners
- **People:** Stein; Peter D'Eustachio; Henning Hermjakob; Guanming Wu and others.
- **Concrete details:** Reactome is an open, peer-reviewed database of molecular pathways; Stein is listed as PI and curator lead
- **Why it matters:** Provides systems-level views of metabolism, signaling, DNA replication, and more — a direct bridge to intro biochemistry teaching.
- **Evidence quality:** Established database documentation.
- **Myth risk:** *Low* — well documented
- **Sources:**
  - https://en.wikipedia.org/wiki/Reactome?utm_source=chatgpt.com
  - https://reactome.org/about/team?utm_source=chatgpt.com

SCENE #5: *Keynote Prediction about the Future of Bioinformatics*

- **Summary**: Provocative O'Reilly Bioinformatics Technology Conference keynote (2003) titled *"Bioinformatics: Gone by 2012."*
- **Date/Place**: 2003; San Diego, Bioinformatics Technology Conference.
- **People**: Stein; audience of early bioinformatics developers and users.
- **Concrete details**: Stein argued that bioinformatics would be absorbed into mainstream biology (tools become ubiquitous)
- **Why it matter**s: Reveals his vision regarding the maturation of the field and the role of computational tools in biology.
- **Evidence quality**: Secondary report of keynote (no full transcript).
- **Myth risk**: *Medium* — known widely in the field but specific quotes vary by retelling.
- Sources:
  - chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://courses.cs.duke.edu/cps006g/fall04/papers/bioinformaticsdead.pdf?utm_source=chatgpt.com

(Repeat for 3–6 scenes.)

2) Direct Quotes From the Scientist (Target 3–7)

Find quotes said or written by the scientist (not quotes about them). Prioritize quotes that reveal how they think or who they are.

For each quote:

1. **Quote:** "I slipped into bioinformatics through the back door…"
   a. **Context/date:** Retrospective essay about career path (The Scientist / online article).

   b. **Source:**https://scienceblogs.com/digitalbio/2007/06/19/to-bioinformatics-students-eve?utm_source=chatgpt.com

2. **Quote:** (paraphrase) “Bioinformatics … is biology” (projected field integration).
   a. **Context/date:** 2003 keynote summary titled *Bioinformatics: Gone by 2012.*
   b. **Source:**chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://courses.cs.duke.edu/cps006g/fall04/papers/bioinformaticsdead.pdf?utm_source=chatgpt.com

3. **Quote (reported):** Stein on Reactome: “…each group does curation, database and quarterly releases … we work collaboratively…”
   a. **Context/date:** Q&A with faculty spotlight.
   b. **Source:**https://moleculargenetics.utoronto.ca/news/medgen-faculty-spotlight-dr-lincoln-stein?utm_source=chatgpt.com

3) Human Reality Details (Collect 8–15, Source-Backed)

- Stein trained as both a physician and a scientist, earning an MD-PhD at Harvard, which shaped his lifelong effort to bridge clinical medicine with computational biology — Wikipedia contributors, *Lincoln Stein*. https://en.wikipedia.org/wiki/Lincoln_Stein
- During medical school and residency, he supported himself by writing software, including programs for radiology and ultrasound analysis, reinforcing a habit of solving biological problems through code — University of Toronto, *MedGen Faculty Spotlight: Dr. Lincoln Stein*. https://moleculargenetics.utoronto.ca/news/medgen-faculty-spotlight-dr-lincoln-stein
- He completed a pathology residency at Brigham and Women’s Hospital but became frustrated by the difficulty of integrating computing into traditional clinical research — Stein, “To bioinformatics students, everywhere,” *ScienceBlogs* (2007). https://scienceblogs.com/digitalbio/2007/06/19/to-bioinformatics-students-eve
- Stein described his entry into bioinformatics as “through the back door,” emphasizing that his career shift was driven by curiosity and dissatisfaction rather than a planned trajectory — Stein (2007), *ScienceBlogs*. https://scienceblogs.com/digitalbio/2007/06/19/to-bioinformatics-students-eve
- While still a pathology resident, he took a summer position with the software group at the Whitehead Institute, a short-term decision that permanently redirected his career — Vector Institute, *Lincoln Stein*. https://vectorinstitute.ai/team/lincoln-stein/
- He postponed and ultimately moved away from completing a clinical fellowship in order to pursue bioinformatics full-time, a risky decision at the time given the field’s uncertain status — University of Toronto, *MedGen Faculty Spotlight*. https://moleculargenetics.utoronto.ca/news/medgen-faculty-spotlight-dr-lincoln-stein
- Stein is known for valuing collaboration and community-driven science, emphasizing shared infrastructure over individual credit in projects like WormBase and Reactome — Bioinformatics Community Conference, *Speaker Profile: Lincoln Stein*. https://bcc2020.sched.com/speaker/lincoln.stein
- He helped create CGI.pm, a foundational Perl module that enabled scientists with minimal programming experience to publish biological data on the web, reflecting his commitment to accessibility — Life Sciences Society, *CSB 2006 Speaker Page*. https://www.lifesciencessociety.org/CSB2006/Stein.html
- Stein has openly stated that most research ideas fail and that failure is an expected and valuable part of scientific progress, shaping a mentoring philosophy tolerant of experimentation — Marathon of Hope

Cancer Centres Network, *Q&A with Dr. Lincoln Stein*. https://www.marathonofhopecancercentres.ca/impact/news-item/meet-the-team-canada-of-cancer-research--a-q-a-with-dr.-lincoln-stein

- He views "eureka moments" as rare and overemphasized, instead describing scientific work as incremental problem-solving — Marathon of Hope Cancer Centres Network, *Q&A with Dr. Lincoln Stein*. https://www.marathonofhopecancercentres.ca/impact/news-item/meet-the-team-canada-of-cancer-research--a-q-a-with-dr.-lincoln-stein
- Stein argued publicly that bioinformatics should eventually disappear as a named discipline because computation would become inseparable from biology itself — Stein, *Bioinformatics: Gone by 2012* keynote summary (2003). https://courses.cs.duke.edu/cps006g/fall04/papers/bioinformaticsdead.pdf
- His leadership style in large consortium projects emphasizes structured curation, regular data releases, and distributed responsibility rather than centralized control — University of Toronto, *MedGen Faculty Spotlight*. https://moleculargenetics.utoronto.ca/news/medgen-faculty-spotlight-dr-lincoln-stein

4) Science Connection to Intro Biochemistry (BIOL 3030 Bridge)

- **Best-fit topic:** *DNA/RNA and Genome-Scale Data Integration*
- **Precise contribution:** Stein helped create foundational databases (WormBase, physical genome maps) and Reactome, enabling structured views of pathways and gene interactions.
    - https://en.wikipedia.org/wiki/Reactome?utm_source=chatgpt.com
- **Bridge explanation (2–4 sentences):**
  Biochemistry studies how molecules like proteins and DNA work inside cells. To understand complex systems like metabolism or signaling, scientists need organized maps of reactions and gene functions. Stein's work on Reactome provides a curated, visual catalogue of biochemical pathways — lines connecting enzymes, substrates, and products — making large datasets interpretable for researchers and students alike.
- **Scientist's own framing:** In his bioinformatics keynote, Stein argued bioinformatics would become *just biology* as computational tools become standard practice in experimental science.
    - chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://courses.cs.duke.edu/cps006g/fall04/papers/bioinformaticsdead.pdf?utm_source=chatgpt.com

5) Complications, Disputes, or Controversies (Only if Evidence Exists)

Item 1: *"Bioinformatics: Gone by 2012" — Debate Over the Identity of the Field*

What's agreed upon:
Stein delivered a widely cited 2003 keynote arguing that bioinformatics would eventually cease to exist as a distinct discipline because computational methods would become fully integrated into mainstream biology.

What's disputed:
Some bioinformaticians and computational biologists disagreed, arguing that the field did not "disappear" but instead expanded and professionalized, remaining a distinct area with its own methods, training pipelines, and departments.

Best sources on both sides:

- Stein, L. (2003). *Bioinformatics: Gone by 2012* (conference keynote summary). https://courses.cs.duke.edu/cps006g/fall04/papers/bioinformaticsdead.pdf
- Stein, L. D. (2008). Towards a cyberinfrastructure for the biological sciences. *Nature Reviews Genetics, 9*(9), 678–688. https://pmc.ncbi.nlm.nih.gov/articles/PMC2646289/
- University of Toronto. *MedGen Faculty Spotlight: Dr. Lincoln Stein*. https://moleculargenetics.utoronto.ca/news/medgen-faculty-spotlight-dr-lincoln-stein

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Item 2: *Credit and Visibility in Large Collaborative Databases*

What's agreed upon:
Projects Stein led or co-led (e.g., WormBase, Reactome) rely on large, distributed collaborations in which individual contributions are intentionally de-emphasized in favor of shared infrastructure and community governance.

What's disputed:
Some critics of big-science biology argue that such models make individual intellectual contributions harder to recognize, particularly for early-career scientists, though this critique is aimed at consortium science broadly rather than Stein personally.

Best sources on both sides:

- Bioinformatics Community Conference. *Speaker profile: Lincoln Stein*. https://bcc2020.sched.com/speaker/lincoln.stein
- Stein, L. D. (2008). *Nature Reviews Genetics*. https://pmc.ncbi.nlm.nih.gov/articles/PMC2646289/

Item 3: *Institutional Risk of Leaving Clinical Medicine*

What's agreed upon:
Stein delayed and ultimately moved away from completing a traditional clinical fellowship to pursue bioinformatics, at a time when computational biology lacked stable career pathways.

What's disputed:
There is no dispute about the facts, but this decision reflects a broader structural issue: institutions were not well prepared to support hybrid clinician–computational scientists in the 1990s.

Best sources:

- Stein, L. (2007). *To bioinformatics students, everywhere*. ScienceBlogs. https://scienceblogs.com/digitalbio/2007/06/19/to-bioinformatics-students-eve
- University of Toronto. *MedGen Faculty Spotlight: Dr. Lincoln Stein*. https://moleculargenetics.utoronto.ca/news/medgen-faculty-spotlight-dr-lincoln-stein

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Item 4: *Curation vs. Automation in Biological Databases*

What's agreed upon:
Stein has consistently advocated for expert human curation in biological databases (e.g., Reactome), even as automated methods improve.

What's disputed:
Some in the field argue that large-scale automation and machine learning should replace or minimize manual curation due to cost and scalability concerns.

Best sources on both sides:

- Reactome contributors. *Reactome pathway database*. https://reactome.org/
- Stein, L. D. (2008). *Nature Reviews Genetics*. https://pmc.ncbi.nlm.nih.gov/articles/PMC2646289/
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6) "Not on Wikipedia" Gold (Aim for 5+ items)

- **Stein paid his way through medical training by writing custom scientific software, including programs for radiology and ultrasound analysis** — This reveals early financial and practical pressure that shaped his habit of solving problems through code, predating formal bioinformatics — University of Toronto, *MedGen Faculty Spotlight: Dr. Lincoln Stein*. https://moleculargenetics.utoronto.ca/news/medgen-faculty-spotlight-dr-lincoln-stein
- **He described his entry into bioinformatics as unplanned and driven by dissatisfaction rather than ambition ("through the back door")** — This undermines the common "visionary planner" trope and shows a reactive, problem-driven career path — Stein, "To bioinformatics students, everywhere," *ScienceBlogs* (2007). https://scienceblogs.com/digitalbio/2007/06/19/to-bioinformatics-students-eve
- **Stein postponed and ultimately stepped away from completing a clinical fellowship to pursue computational biology full-time** — This highlights a significant professional risk taken when bioinformatics had little institutional legitimacy — University of Toronto, *MedGen Faculty Spotlight*. https://moleculargenetics.utoronto.ca/news/medgen-faculty-spotlight-dr-lincoln-stein
- **He was a principal developer of CGI.pm, a foundational Perl module that enabled scientists with limited programming skills to publish biological data on the early web** — This contribution sits at the intersection of web history and biology and is often omitted from biology-focused biographies — Life Sciences Society, *CSB 2006 Speaker Page*. https://www.lifesciencessociety.org/CSB2006/Stein.html
- **Stein explicitly rejected the idea of "eureka moments," emphasizing instead that most research ideas fail and that failure is routine and productive** — This provides rare insight into his mentoring philosophy and working mindset — Marathon of Hope Cancer Centres Network, *Q&A with Dr. Lincoln Stein*. https://www.marathonofhopecancercentres.ca/impact/news-item/meet-the-team-canada-of-cancer-research--a-q-a-with-dr.-lincoln-stein
- **He argued that bioinformatics should eventually become invisible because it would be absorbed into everyday biological practice** — This philosophical stance shaped his emphasis on infrastructure over personal visibility and sparked long-running debate in the field — Stein, *Bioinformatics: Gone by 2012* (2003 keynote summary). https://courses.cs.duke.edu/cps006g/fall04/papers/bioinformaticsdead.pdf
- **In large projects like Reactome, Stein emphasized distributed curation and regular releases rather than centralized control** — This governance philosophy helps explain the durability and

adoption of the database but is rarely highlighted in summaries — University of Toronto, *MedGen Faculty Spotlight*.
https://moleculargenetics.utoronto.ca/news/medgen-faculty-spotlight-dr-lincoln-stein

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Required Sources

Minimum 6 sources, preferably 10+. Prioritize:

1. Stein's professional profiles: vectorinstitute.ai; bioinformatics.ca; oicr.
    a. https://vectorinstitute.ai/team/lincoln-stein/?utm_source=chatgpt.com
    b. https://bioinformatics.ca/instructors/lincoln-stein/?utm_source=chatgpt.com
    c. https://oicr.on.ca/researchers/dr-lincoln-stein/?utm_source=chatgpt.com
2. Wikipedia & HandWiki summaries.
    a. https://en.wikipedia.org/wiki/Lincoln_Stein?utm_source=chatgpt.com
    b. https://handwiki.org/wiki/Biography%3ALincoln_Stein?utm_source=chatgpt.com

3. Retrospective essays and interviews (ScienceBlogs, Marathon of Hope interview).
    a. https://scienceblogs.com/digitalbio/2007/06/19/to-bioinformatics-students-eve?utm_source=chatgpt.com
    b. https://www.marathonofhopecancercentres.ca/impact/news-item/meet-the-team-canada-of-cancer-research--a-q-a-with-dr.-lincoln-stein?utm_source=chatgpt.com
4. PubMed articles and database documentation (Bioinformatics editorial, Reactome).
    a. https://pmc.ncbi.nlm.nih.gov/articles/PMC2646289/?utm_source=chatgpt.com
    b. https://en.wikipedia.org/wiki/Reactome?utm_source=chatgpt.com

5. Faculty spotlight narrative with personal details.
    a. https://moleculargenetics.utoronto.ca/news/medgen-faculty-spotlight-dr-lincoln-stein?utm_source=chatgpt.com

6. Secondary conference reports.
    a. chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://courses.cs.duke.edu/cps006g/fall04/papers/bioinformaticsdead.pdf?utm_source=chatgpt.com

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Output Format (Strict)

A) ANCHOR SCENES (3–6)

- **SCENE #1:**
    - Summary:
    - Date/Place:
    - People:
    - Concrete details:
    - Why it matters:
    - Evidence quality:

    - Myth risk:
    - Sources:
      (Repeat)

B) DIRECT QUOTES (3–7)

1. "…"
    - Context/date:
    - Source (full citation):
      (Repeat)

C) HUMAN REALITY DETAILS (8–15 bullets)

- … — Source:
  (Repeat)

D) SCIENCE CONNECTION (Intro Biochem)

- Best-fit topic:
- Precise contribution (1 sentence):
- Bridge explanation (2–4 factual sentences):
- Scientist's own framing/analogy (if found):
- Sources:

E) COMPLICATIONS OR CONTROVERSIES (If any)

- Item:
    - What's agreed:
    - What's disputed:
    - Sources:

F) "NOT ON WIKIPEDIA" FINDS (5+)

- Finding — Why it's distinctive — Source:

G) SOURCES CONSULTED (Full citations)

1.
2.

…

H) FLAGGED "MYTH-RISK" DETAILS (If any)

List any dramatic details that are frequently repeated but weakly sourced.

- Claim:
- Where it appears:
- Why sourcing is weak:
- Best available evidence (if any):

LENS RECOMMENDATION (Do this only after completing A–G)

Lens scoring rubric (0–3 each)

Score based on how well your collected evidence supports a scene-driven narrative.

Eureka (Discovery crystallized)

- 3: Documented discovery moment with concrete details and strong sourcing
- 2: Clear discovery story but details thin or second-hand
- 1: Only broad “discovered X” claims
- 0: No identifiable discovery moment

Failure (Setback that changed trajectory)

- 3: Specific, well-documented setback that clearly shapes what happens next
- 2: Struggle exists but not clearly pivotal
- 1: Generic “it was hard”
- 0: None

Mentor (Relationship that redirected them)

- 3: Specific mentorship interaction or relationship pivot with clear before/after
- 2: Mentorship exists but pivotal moment is fuzzy
- 1: Only “worked with…”
- 0: None

Controversy (Conflict over credit, ideas, ethics, institutions)

- 3: Clear dispute with parties, stakes, and outcome; preferably multiple perspectives
- 2: Tension exists but one-sided or unclear
- 1: Vague conflict claims
- 0: None

Human Detail (Revealing personal fact that changes how we read the science)

- 3: Well-sourced detail that illuminates character and connects back to their work
- 2: Interesting but not connected to the scientific story
- 1: Trivia
- 0: None

Last Experiment (Final act, handoff, or late-life scene with closure)

- 3: Strong late-career or final-scene moment with concrete details and good sourcing
- 2: Late career story exists but not scene-worthy
- 1: Only death/retirement facts
- 0: None

Decision rule

- Choose the lens with the highest score.
- If tied, choose the lens tied to the most vivid, best-sourced anchor scene.
- Provide one backup lens only if it is within 1 point of the top score.

Output for lens recommendation

- Score table (lens: score 0–3 + 1 sentence justification)
- Primary recommended lens:
    - Why it fits (cite which scene card numbers support it)
    - Best opening scene (scene #)
    - One quote to use early (quote #)
    - Cautions (thin sourcing, disputed detail, myth risk)
- Backup lens (if applicable):
    - Same items, briefly

Final Note

Do not write the profile. Deliver only evidence, citations, and lens recommendation after evidence collection.

LENS SCORING TABLE

Lens	Score (0–3)	Justification
Eureka (Discovery crystallized)	**1**	Stein's impact comes from infrastructure and integration rather than a single, documentable discovery moment; no sharp "aha" scene with primary documentation.
Failure (Setback that changed trajectory)	**3**	Well-documented career redirection away from clinical medicine/pathology toward bioinformatics, explicitly described by Stein as driven by frustration and dissatisfaction (Scenes #1, #5).
Mentor (Relationship that redirected them)	**1**	Collaborations (e.g., Whitehead Institute, consortium science) are present, but no single mentor relationship with a clear before/after pivot is documented.
Controversy (Conflict over ideas/institutions)	**2**	Field-level controversy over whether bioinformatics should exist as a discipline ("Gone by 2012"), but no personal dispute or high-stakes conflict with named adversaries.

Human Detail (Revealing personal fact)	**3**	Strong, well-sourced personal details (self-funded training via software, rejection of “eureka” myths, tolerance for failure) that directly explain how and why Stein built shared infrastructure.
Last Experiment (Late-career closure)	**1**	Active career with no final handoff or closing scene; late-career philosophy exists but is not scene-driven.

PRIMARY RECOMMENDED LENS

Failure → Redirection → Infrastructure Builder

Why it fits:
This lens is best supported by **specific, well-documented setbacks** and **career dissatisfaction** that directly caused Stein’s pivot into bioinformatics and shaped his emphasis on invisible, shared scientific infrastructure rather than individual discovery.

- **Supporting scenes:**
    - **Scene #1** (Entering bioinformatics “through the back door”)
    - **Scene #5** (2003 keynote reframing the field’s future)

Best opening scene:

- **Scene #1** — the summer at the Whitehead Institute while still a pathology resident, when frustration with traditional clinical research catalyzed a permanent career shift.

One quote to use early:

- **Quote #1:** “I slipped into bioinformatics through the back door…” (Stein, 2007)

Cautions:

- Avoid overstating this as “failure” in a dramatic sense; it is better framed as **structural frustration and institutional mismatch**, not incompetence or collapse.
- The “back door” phrasing is memorable but retrospective; treat it as reflective, not contemporaneous documentation.

BACKUP LENS (Within 1 Point)

Human Detail → Philosophy of Invisible Science

Why it fits:
Stein's documented rejection of "eureka moments," comfort with failure, and commitment to community-driven tools provide a strong interpretive frame for understanding why he prioritized databases and infrastructure over headline discoveries.

Supporting scenes:

- **Scene #3** (WormBase)
- **Scene #4** (Reactome)

Best opening scene:

- **Scene #5** — the "Bioinformatics: Gone by 2012" keynote as a philosophical declaration.

Quote to use:

- **Quote #2** (Bioinformatics will become biology)

Cautions:

- Risk of sounding abstract if not grounded in concrete scenes (must anchor philosophy to database-building work).
- The keynote is secondarily sourced; quote carefully.